DSA EXPERIMENT 6: Stack using Linked List.

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BATCH:B1 **CLASS:SY1**

Stack using linked list

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
   int data;
   struct Node *next;
};

struct Node *top = NULL;

void push(int x)
{
   struct Node *newnode;

   newnode = malloc(sizeof(struct Node));

   newnode->data = x;
   newnode->next = top;
   top = newnode;
}

void pop()
{
   struct Node *temp;

   if (top == NULL)
      printf("Stack is empty\n");
   else
   {
      temp = top;
      printf("Popped element: %d\n", top->data);
      top = top->next;
      free(temp);
   }
}
```

```
void display()
{
    struct Node *temp = top;

    if (top == NULL)
        printf("Stack is empty\n");
    else
    {
        while (temp != NULL)
        {
            printf("%d ", temp->data);
            temp = temp->next;
        }
        printf("\n");
    }
}

int main()
{
    push(10);
    push(20);
    push(30);

    display();

    pop();

    display();

    return 0;
}
```

Output

```
30 20 10
Popped element: 30
20 10


=== Code Execution Successful ===
```